

Aaron Lee Engineering Portfolio - AI Handoff Document

Document purpose: Structured review of all user requests, changes made, current project state, and operational commands. Intended for another AI model to resume work without prior conversation context.

Generated: 2026-05-30

Repository: /Users/alee/Desktop/Website0.1

GitHub remote: <https://github.com/aaronlee5702-dot/Website0.1.git>

Live domain (per user): aaronwlee.com (Vercel deployment)

Latest commit at generation: 3dca109 - "Add profile photo to hero section"

1. Project Overview

Field	Value
Project name	Aaron Lee Engineering Portfolio
Framework	Next.js 16.x (App Router)
Language	TypeScript
Styling	Tailwind CSS 4
Icons	lucide-react
Analytics	@vercel/analytics (enabled on Vercel dashboar
Hosting	Vercel
Dev command	npm run dev (uses Turbopack)
Build command	npm run build
Lint command	npm run lint

Primary data source

All portfolio content lives in `src/data/portfolio.ts`. Components read from this file; most content edits should happen there.

Key routes

Route	File	Purpose
/	<code>src/app/page.tsx</code>	Home (Hero, About, Skills, Projects, Experien
/all-projects	<code>src/app/all-projects/page.tsx</code>	Full project grid with category filters
/projects/[id]	<code>src/app/projects/[id]/page.tsx</code>	Individual project detail pages
/soldering-projects	<code>src/app/soldering-projects/page.tsx</code>	Subfolder for soldering projects
/resume.pdf	<code>public/resume.pdf</code>	Resume (linked from Header)

2. Architecture Notes for AI Assistants

Content model (portfolio.ts)

- `projects[]` - Main project cards (ids '1'-'9' currently)
- `solderingProjects[]` - Separate array for Bonus Soldering Projects subfolder (currently empty)
- `experience[]`, `education[]`, `skills[]`, `personalInfo`

UI patterns established by user

1. Aurora borealis gradient - CSS classes `aurora-bg` and `aurora-content` in `globals.css`; used on project card headers and page

headers.

2. Home Featured Projects - Shows only first 4 projects (projects.slice(0, 4)). No category filter bar on home; replaced with Soldering bonus card linking to /soldering-projects.
3. All Projects page - Keeps category filters (All / Design / Analysis / Manufacturing / Research) plus a special "Bonus Soldering Projects" card.
4. Skills section - Category tabs show 2 related project cards (not percentage bars). "Skills Overview" at bottom shows skill names as pills when category clicked. Manufacturing & Processes includes Bonus -> Soldering link.
5. Project detail pages - Description from detailedDescription; gallery from images, videos, pdfs arrays. "View Source Code" button was removed per user request.
6. Contact section - Form and "Current Availability" removed; only contact info remains.
7. Header - Logo text: "Aaron Lee Portfolio". Resume opens /resume.pdf in new tab.
8. Hero profile photo - public/profile.png with CSS background positioning in Hero.tsx (backgroundSize: '211%', backgroundPosition: '45% 55%').

Files most frequently edited

```
src/data/portfolio.ts
src/components/Hero.tsx
src/components/About.tsx
src/components/Projects.tsx
src/components/Skills.tsx
src/components/Contact.tsx
src/components/Header.tsx
src/components/Experience.tsx
src/app/all-projects/page.tsx
src/app/projects/[id]/page.tsx
src/app/soldering-projects/page.tsx
src/app/layout.tsx
```

3. Complete Chronological User Request Log

Each entry: User request -> Action taken -> Status

Phase A - Initial setup & About/Skills/Contact

#	User request	Implementation	Status
1	Run on local dev server	npm install, npm run dev	Done
2	About education: replace achievement	Updated education in portfolio.ts	Done
3	Remove "Renewable Energy Research"	Updated About.tsx	Done
4	Skills: PLC Programming -> Java, La	Updated skills in portfolio.ts	Done
5	Contact: remove "Send a Message" a	Updated Contact.tsx	Done
6	Remove "Get In Touch" header/subtit	Updated Contact.tsx	Done

Phase B - Project descriptions & content

#	User request	Implementation	Status
7	Add Description section on project de	Added detailedDescription field + UI i	Done
8	6-axis arm: remove testing sentence;	Updated project id '1'	Done (later replaced with resume text)
9	Automated Lab System: remove ANS	Updated project id '2'	Done

10	Add bullet points for incubator project	Updated detailedDescription for id '2'	Done
11	Replace heat exchanger sentence with	Updated brief description	Done
12	Add SolidWorks FEA and CAD to incubator	Updated technologies array	Done
13	Prusa MK3s: change brief description	Updated project id '3'	Done
14	Prusa MK3s: bullet points in description	Updated detailedDescription	Done
15	Wind Turbine: remove patent, change	Updated highlights	Done
16	Prusa: add enclosure & smaller nozzle	Updated detailedDescription	Done
17	Wind Turbine: replace final sentence	Updated detailedDescription	Done
18	Wind Turbine: add SolidWorks FEA and	Updated technologies	Done
19	Hero: "3 Patents Filed" -> "2 Research	Updated Hero.tsx	Done
20	Update LinkedIn link	Updated personalInfo.linkedinin	Done
21	Fix GitHub link (was wrong URL)	Updated personalInfo.github to aaron	Done

Phase C - About, branding, deployment

#	User request	Implementation	Status
22	About: replace mentoring/open-source	Updated About.tsx	Done
23	About: 6 years mechanical -> 6 years	Updated About.tsx	Done
24	About: "business value" -> "advanced	Updated About.tsx	Done
25	Deploy with Vercel	Vercel CLI login & deploy	Done
26	TSMC title -> full company name	Updated experience entry	Done
27	TSMC location -> Phoenix, AZ; dates	Updated experience	Done
28	AleePort -> Aaron Lee Portfolio	Updated Header.tsx	Done
29	Site title -> Aaron Lee Engineering Po	Updated layout.tsx metadata	Done
30	Add iRhythm Technologies experience	Added as experience id '1', shifted ot	Done
31	iRhythm description update & location	Updated experience	Done
32	Replace browser favicon with UC Ber	Attempted; corrupted favicon remove	Reverted/failed - favicon removed
33	Fix resume access	Resume at public/resume.pdf, Heade	Done
34	Hero: "15+ Technologies" -> "15+ En	Updated Hero.tsx	Done
35	Contact intro text edit (remove consu	Updated Contact.tsx	Done

Phase D - Projects UI & new pages

#	User request	Implementation	Status
36	"More Projects" button -> All Projects	Created src/app/all-projects/page.tsx	Done
37	Aurora animation on All Projects head	Applied aurora-bg classes	Done
38	Move "All Projects" button below sub	Updated Projects.tsx	Done
39	Aurora effect on all project cards (hor	Updated card headers	Done
40	Home page: only first 4 projects	projects.slice(0, 4)	Done
41	Add SEB: Tracking Project	Added project id '5'	Done
42	Trim SEB description (remove last 2 s	Updated detailedDescription	Done
43	Push to GitHub	Committed & pushed	Done
44	Next.js security vulnerability (Vercel c	npm install next@latest, audit fix	Done
45	6-axis arm description from resume P	Replaced detailedDescription	Done

46	Automated lab description from resur	Replaced detailedDescription	Done
47	Display incubator pictures on project	Added images/videos arrays	Done

Phase E - Skills redesign & new projects

#	User request	Implementation	Status
48	Skills: show 2 related project cards in	Redesigned Skills.tsx	Done
49	Skills Overview: clickable categories,	Updated Skills.tsx	Done
50	Add Soldering to Manufacturing skills	Added to skills array	Done
51	Add Self Balancing Cube & DIY Gimbal	Projects id '7', '8'	Done - awaiting content
52	Bonus Soldering Projects subfolder c	solderingProjects array + /soldering-p	Done
53	Manufacturing skills: Bonus Soldering	Added in Skills.tsx	Done
54	Home Featured Projects: replace filter	Updated Projects.tsx	Done
55	Add Plant Growing Hydroponic System	Project id '9' (empty details)	Done - awaiting content

Phase F - Analytics, git, profile photo

#	User request	Implementation	Status
56	Implement Vercel Web Analytics	npm install @vercel/analytics, <Analy	Done
57	Explain git push commands	Provided git add, commit, push instru	Done
58	Terminal: how to exit dev server	Explained Ctrl+C or new terminal tab	Done
59	Analytics already working on Vercel	Confirmed enabled (1 visitor, 2 page	Done
60	Explain Visitors / Page Views / Bounc	Provided definitions	Done
61	Upload profile photo to Hero circle	public/profile.png, updated Hero.tsx	Done
62	Fix photo framing (LinkedIn reference	CSS background-size/position tuning	Done
63	Zoom adjustments (5%, 10%, 2%, re	Final: backgroundSize: '211%'	Done
64	Push profile photo to GitHub	Commit 3dca109 pushed to main	Done

4. Current Project Inventory

Main projects (projects[])

ID	Title	Category	Content status
1	6-axis Robotic Arm	design	Complete
2	Automated Lab System Incubator	research	Complete + gallery
3	Additive Manufacturing: Prusa MK3s	manufacturing	Complete
4	Wind Turbine Blade Aerodynamics G	analysis	Complete
5	SEB: Tracking Project	analysis	Complete + gallery
6	Portable Microscope	manufacturing	Complete + gallery
7	Self Balancing Cube	design	Empty - user will fill later
8	DIY Gimbal	design	Empty - user will fill later
9	Plant Growing Hydroponic System	design	Empty - user will fill later

Soldering subfolder (solderingProjects[])

- Currently empty. Add entries here; they appear on /soldering-projects only.

Professional experience (order matters - most recent first)

1. iRhythm Technologies - Tools & Fixtures System Engineer - Jan 2026 - Present - San Francisco, CA
2. Tsao Lab: NeuroVision - Mechanical Design Engineer - 2025 - Present - Berkeley, CA
3. TSMC - Epitaxy Equipment Engineer Intern - May 2025 - Aug 2025 - Phoenix, AZ

Personal info

- Name: Aaron Lee
- Title: B.S. Mechanical Engineer
- Email: oisaaronlee@gmail.com
- Phone: +1 (925) 885-9137
- Location: San Francisco, CA
- LinkedIn: <https://www.linkedin.com/in/aaronlee5702/>
- GitHub: <https://github.com/aaronlee5702-dot>

Hero statistics (hardcoded in Hero.tsx)

- 4+ Years CAD Experience
- 10+ Projects Completed
- 15+ Engineering Skills
- 2 Research Projects

5. Git Commit History (Reference)

3dca109	2026-05-30	Add profile photo to hero section
4e64703	2026-05-30	update projects and soldering systems
0bbecaa	2026-01-31	Update 6-axis robotic arm and automated lab system descriptions
46638b2	2026-01-31	Update Next.js to fix security vulnerability
ff1ff1b	2026-01-31	Add iRhythm Technologies experience and update location
6a5e66e	2025-11-12	resume changes
646b5f6	2025-11-10	Fix: Remove unused ArrowLeft import causing Vercel build error
464f2a0	2025-11-10	detail changes
7e70e7e	2025-11-10	changed the icon
a0a6a31	2025-11-10	added project photos and a new card
3339bd4	2025-10-17	SEB tracking
c2709c8	2025-10-15	Add SEB Tracking Project and limit home page to 4 featured projects
0dc381c	2025-10-12	Add aurora borealis effects and All Projects page
2d70717	2025-10-02	Tsao Lab, TSMC, About, project descriptions
54b6921	2025-09-03	first commit

6. Shell Commands Reference (User & Assistant)

Development

```
cd /Users/alee/Desktop/Website0.1
npm install          # Install dependencies
npm run dev          # Start dev server (Turbopack) - blocks terminal
```

```
npm run build      # Production build
npm run lint       # ESLint
```

Git workflow (user standard)

```
git add .
git commit -m "Your message here"
git push
```

Stop dev server

- Press Ctrl + C in the terminal running npm run dev
- Or open a new terminal tab (Ctrl + Shift +) to run git commands while server runs

Vercel deployment

- Push to main triggers automatic Vercel deploy
- Manual: npx vercel --prod (requires vercel login)

Analytics setup (already done)

```
npm install @vercel/analytics
```

Component in src/app/layout.tsx:

```
import { Analytics } from "@vercel/analytics/next";
// Inside <body>: <Analytics />
```

7. Known Issues & Pending Items

Item	Notes
Favicon	Removed due to corrupted 0-byte file. User ma
Projects 7, 8, 9	Placeholder cards with empty description, tec
solderingProjects[]	Empty - subfolder shows "coming soon" empty s
Education data	May still contain placeholder MIT/MS entries
npm audit	8 vulnerabilities reported at last install -
Profile photo framing	Tuned to user preference at 211% zoom - may n

8. How to Continue Work (AI Instructions)

Adding a new main project

1. Add object to projects[] in src/data/portfolio.ts with next sequential id.
2. Optionally map to a skill category in Skills.tsx categoryProjects object.
3. Add assets under public/projects/<project-name>/.
4. Project auto-appears on /all-projects and gets detail page at /projects/<id>.

Adding a soldering project

1. Add to solderingProjects[] (NOT projects[]).

2. Appears on /soldering-projects only.

Updating hero profile photo

- Replace public/profile.png
- Adjust in Hero.tsx: backgroundSize and backgroundPosition

User preferences observed

- Do not commit unless explicitly asked
- Do not push unless explicitly asked
- Match existing code style; minimal diffs
- Use resume PDF as source of truth for project descriptions when requested
- Keep home page to 4 featured projects only
- Preserve aurora styling on cards and headers

9. JSON Machine-Readable Summary

```
{
  "project": {
    "name": "Aaron Lee Engineering Portfolio",
    "repo_path": "/Users/alee/Desktop/Website0.1",
    "github": "https://github.com/aaronlee5702-dot/Website0.1.git",
    "domain": "aaronwlee.com",
    "stack": ["Next.js 16", "TypeScript", "Tailwind CSS 4", "Vercel"],
    "primary_data_file": "src/data/portfolio.ts"
  },
  "latest_commit": "3dca109",
  "pending_content": {
    "projects_awaiting_details": ["7", "8", "9"],
    "soldering_projects_empty": true,
    "favicon_missing": true
  },
  "key_routes": ["/", "/all-projects", "/projects/[id]", "/soldering-projects"],
  "total_user_requests_logged": 64,
  "analytics": {
    "provider": "vercel",
    "package": "@vercel/analytics",
    "enabled_on_vercel": true
  }
}
```

End of AI Handoff Document